



COLLEGE PROJECT PRESENTATION & VIVA

DJ MART

E-Commerce Website

"Smart, Simple & Convenient Online Shopping"

⚡ Developed using Antigravity

PROJECT DETAILS

PROJECT TITLE

DJ MART E-Commerce Platform

STUDENT / PRESENTER

Candidate Name

REGISTER NUMBER

[Register Number]

DEPARTMENT

Dept. of Computer Science & Engineering

INSTITUTION & ACADEMIC YEAR

[Engineering College] | 2025 – 2026

PROJECT OVERVIEW

Introduction

A Modern Online Shopping Platform Designed for Retail Convenience



Online Product Browsing

Effortless catalog navigation enabling customers to discover curated luxury, tech, and everyday items anytime.



Real-Time Product Discovery

Instant keyword search and multi-category filtering allowing shoppers to locate target goods in milliseconds.



Detailed Product Pages

Rich presentation featuring crisp photography, transparent pricing in ₹, live inventory badges, and specs.



Interactive Shopping Cart

Seamless quantity adjustments (+/-), live subtotal recalculations, and item removals without page reloads.



Streamlined Order Process

Frictionless checkout capturing delivery details, payment choice, and safe atomic order placement.



User-Friendly Responsive UI

Distinctive luxury design language built with mobile-first adaptability across phones, tablets, and desktops.

MARKET NEED & CONTEXT

Problem Statement

Overcoming Traditional Retail Bottlenecks with a Digital Platform

⚠️ TRADITIONAL SHOPPING BOTTLENECKS

Mandatory Store Visits: Requires physical commute, parking hassles, traffic, and geographical boundaries.

Time-Consuming Process: Hours wasted traversing aisles, searching shelves, and waiting in billing queues.

Difficult Product Comparison: Hard to compare multiple alternative products, prices, and specifications.

Restricted Store Timings: Inaccessible outside standard business hours; unavailable during holidays and late nights.

Uncertain Stock Availability: Shoppers travel to physical stores only to find target products out of stock.

✅ THE DJ MART DIGITAL SOLUTION

Universal 24/7 Access: Shop from any device at any time, eliminating travel and geographical limits.

Frictionless Fast Shopping: Discover products, update cart, and place orders in under two minutes.

Instant Side-by-Side Comparison: Clear pricing, high-res photos, and complete specifications.

Always Open & Reliable: Uninterrupted digital storefront operating 365 days a year without downtime.

Live Inventory Transparency: Real-time stock status prevents ordering unavailable products.

SCOPE & ROADMAP

Project Objectives

Key Engineering & Experience Goals Achieved in DJ Mart

01**Develop Modern Web App**

Build a full-featured, secure e-commerce marketplace using Antigravity AI engineering.

02**Easy Product Browsing**

Provide seamless catalog navigation across luxury, tech, and lifestyle merchandise.

03**Real-Time Search**

Implement responsive keyword search with instant query execution.

04**Structured Categories**

Organize catalog into intuitive categories (Electronics, Fashion, Home, Books).

05**Dynamic Shopping Cart**

Support quantity updates (+/-) and price recalculations without page reloads.

06**Frictionless Checkout**

Deliver validated delivery address capture and atomic order placement.

07**Responsive Interface**

Ensure seamless touch navigation across smartphone, tablet, and desktop viewports.

08**Elevate Experience**

Combine luxury editorial aesthetics with high-performance Java backend architecture.

ENGINEERING ARCHITECTURE

Technologies & Tools Used

Verified Development Stack Actually Implemented in DJ Mart

AI PLATFORM & IDE**Google Antigravity***Development Environment*

Advanced agentic AI assistant utilized for full-stack architecture design, Java code generation, testing, and system verification.

FRONTEND LAYER**HTML5, CSS3 & Vanilla JS***Client Interface*

Custom luxury design system with CSS custom properties (variables), native ES6+ Fetch API client, and zero heavy frontend libraries.

VIEW ENGINE**Jakarta Server Pages (JSP)***Dynamic Server Templates*

Jakarta JSP 3.1 & JSTL 3.0 tag library providing clean server rendering, output escaping against XSS, and modular components.

BACKEND ENGINE**Jakarta Servlet 6.0 & Java 17***Controllers & Business Logic*

High-speed Java enterprise web architecture implementing Layered MVC (Servlets, Filter Chain, Services, and DAOs).

DATABASE**H2 Relational Database (SQL)***Embedded Relational Store*

Embedded SQL database engine with ACID transactions, foreign keys, CHECK constraints, and automated schema migration on boot.

SERVER & SECURITY**Tomcat 10.1 & jBCrypt***Deployment & Cryptography*

Apache Tomcat 10.1 container, HikariCP connection pool, salted BCrypt password hashing, and CSRF synchronizer tokens.

TECHNICAL BLUEPRINT

System Architecture

Multi-Tier Layered MVC Design & Complete Request Lifecycle



LAYER RESPONSIBILITIES

- **Presentation Layer (JSP & JS)**: Renders semantic HTML via JSTL and issues asynchronous REST JSON API calls.
- **Security Layer (Filters)**: Verifies UTF-8 encoding, security headers, CSRF synchronizer tokens, and RBAC roles.
- **Controller Layer (Servlets)**: Intercepts HTTP requests, validates inputs, and maps to services or views.
- **Business Logic Layer (Services)**: Enforces stock constraints, discount rules, and transactional rollbacks.
- **Data Access Layer (DAOs)**: Parameterized PreparedStatements ensuring 100% defense against SQL injection.

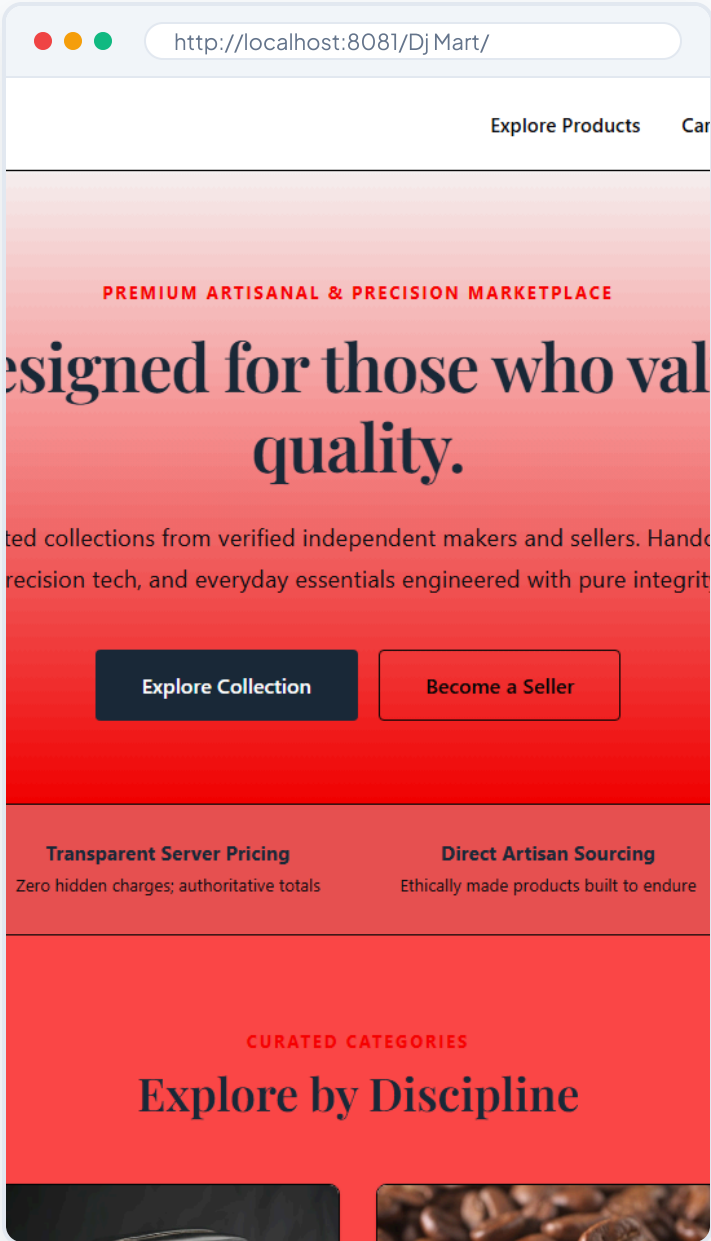
H2 DATABASE ENGINE

- Embedded relational storage (jdbc:h2:./data/djmart)
- Schema: users, products, orders, order_items, cart_items, reviews
- Foreign Keys with ON DELETE CASCADE rules
- ACID transactions on order placement with rollback
- HikariCP connection pool with maximum 10 connections

USER INTERFACE SHOWCASE

DJ Mart – Home Page

Actual Live Screenshot of the DJ Mart Landing & Discovery Experience



01. STICKY LUXURY NAVBAR

Official DJ Mart branding logo, search bar, category shortcuts, live cart count badge, and account menu.

02. EDITORIAL HERO BANNER

"Timeless Elegance, Modern Luxury" showcasing premium lifestyle curation with direct call-to-actions.

03. CATEGORY SHORTCUTS

Instant filter cards for Electronics, Fashion, Home & Kitchen, and Books & Stationery.

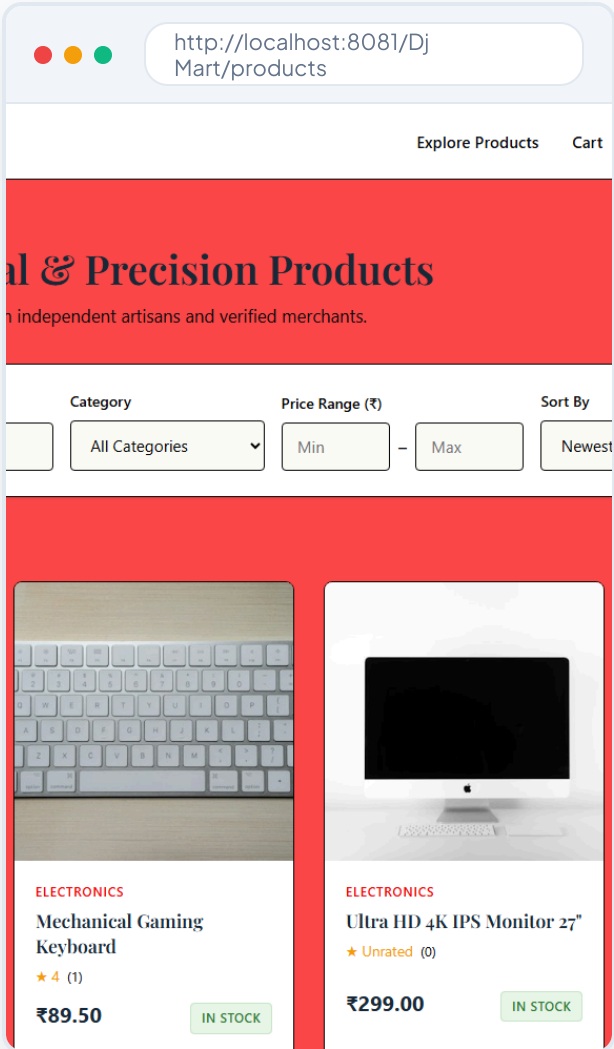
04. FEATURED PRODUCTS GRID

High-res product photography, pricing in ₹, star ratings, and one-click "Add to Cart" triggers.

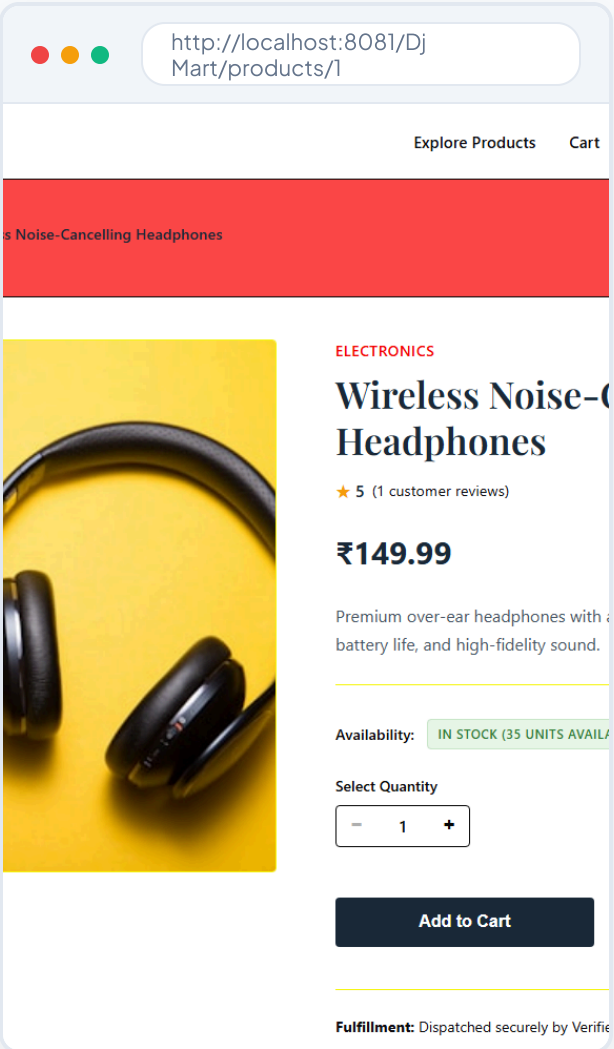
CATALOG & EXPLORATION

Product Browsing & Details

Actual Live Screenshots of Catalog Search, Filters, and Deep Product Insights



CATALOG WITH CATEGORY FILTERS & SORTING



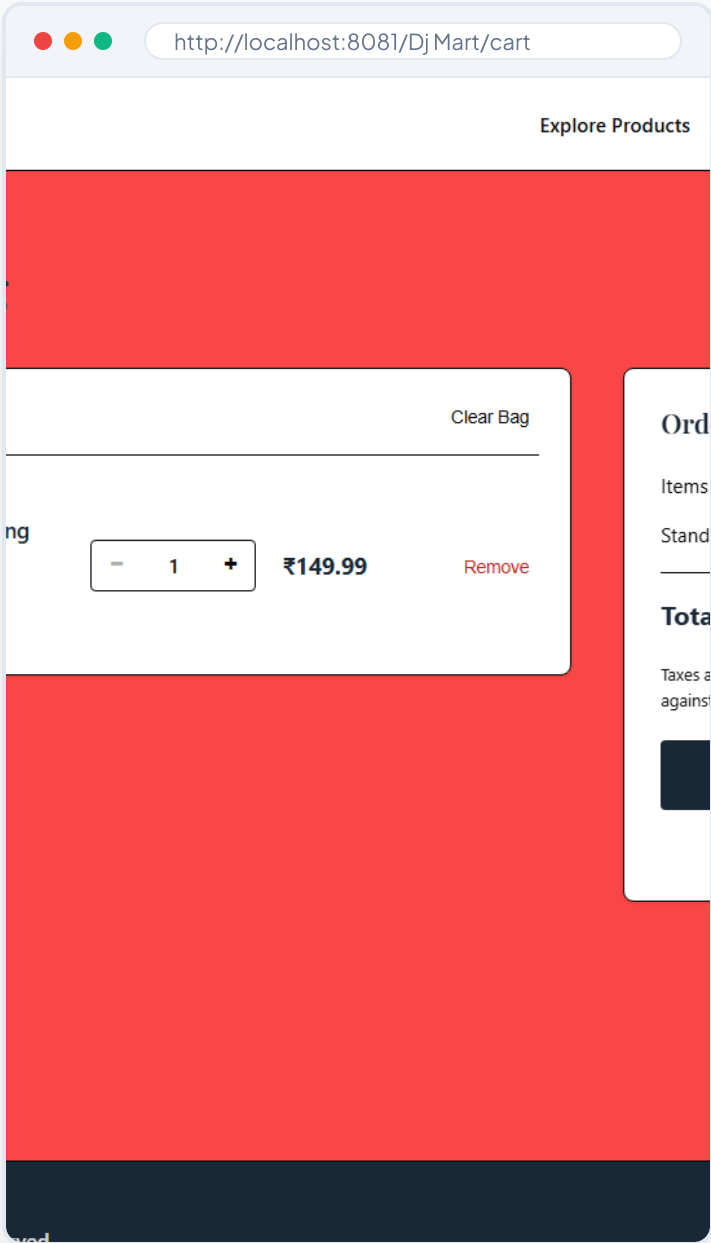
DETAILED PRODUCT VIEW & SPECIFICATIONS

KEY CAPABILITIES: Multi-category filter (Electronics, Fashion, Home) • Price sorting (Low to High / High to Low) • High-res gallery • Live stock indicator badge • Full technical specifications • Direct "Add to Cart" with instant feedback.

ORDER MANAGEMENT

Shopping Cart

Actual Live Screenshot of Interactive Cart with Live Price Recalculation



01. SELECTED PRODUCT LIST

Displays product thumbnail, name, unit price in ₹, and line-item total clearly for verification.

02. DYNAMIC QUANTITY STEPPER

Users can increment (+) or decrement (-) quantities. JavaScript updates /api/v1/cart/items without page reloads.

03. INSTANT ITEM REMOVAL

One-click delete button triggers instant DOM update and adjusts the navbar cart count badge.

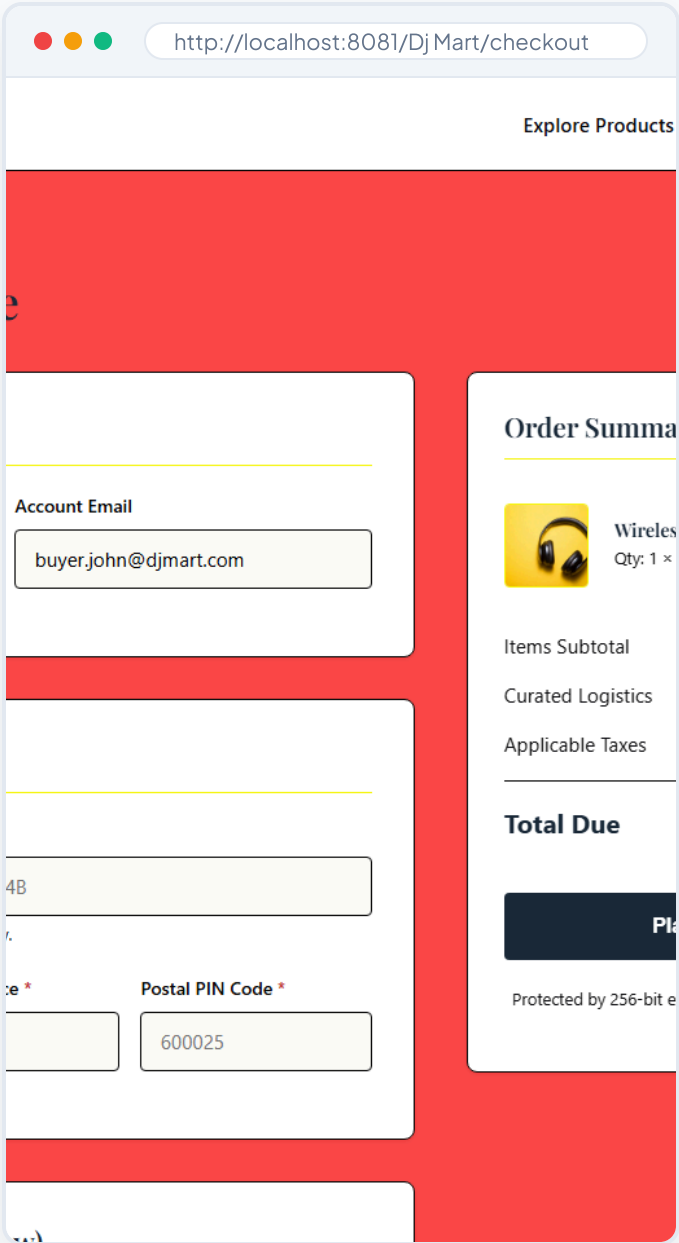
04. ORDER SUMMARY & CHECKOUT

Subtotal, estimated shipping, taxes, and final payable amount leading to the checkout flow.

ORDER FULFILLMENT

Checkout & Order Process

Actual Live Screenshot of Shipping Capture & ACID Transaction Lifecycle



END-TO-END CHECKOUT LIFECYCLE

- 1 Browse Catalog:** Customer explores products and checks specs.
- 2 Add to Cart:** Item placed into session cart via asynchronous API.
- 3 Review Cart:** Quantities verified and totals confirmed by buyer.
- 4 Delivery Details:** Shipping address & contact info validated.
- 5 Payment Selection:** Payment mode selected (Cash on Delivery).
- 6 Place Order & Commit:** Server executes ACID transaction: verifies stock, decrements inventory, creates order record, and clears cart safely.

PLATFORM CAPABILITIES

Key Features of DJ Mart

Robust Set of Verified Features Implemented in Current Release



User-Friendly Luxury UI

Bespoke luxury visual language featuring gold accents, obsidian serif typography, and zero generic styling.



Real-Time Search

Integrated search bar executing instant keyword queries across product titles, categories, and descriptions.



Structured Categories

Multi-tier categorization organizing Electronics, Fashion, Home Essentials, and Stationery.



Detailed Product View

High-res photography, complete specifications, live stock indicators, and price breakdowns.



Dynamic Interactive Cart

AJAX quantity adjustments (+/-), live total recalculation, and single-click item removal with zero reload.



Validated Checkout

Delivery address form validation, payment mode selection, and atomic order placement.



Responsive Design

Fluid layout, slide-out mobile drawer menu, and touch-optimized buttons (minimum 44x44px).



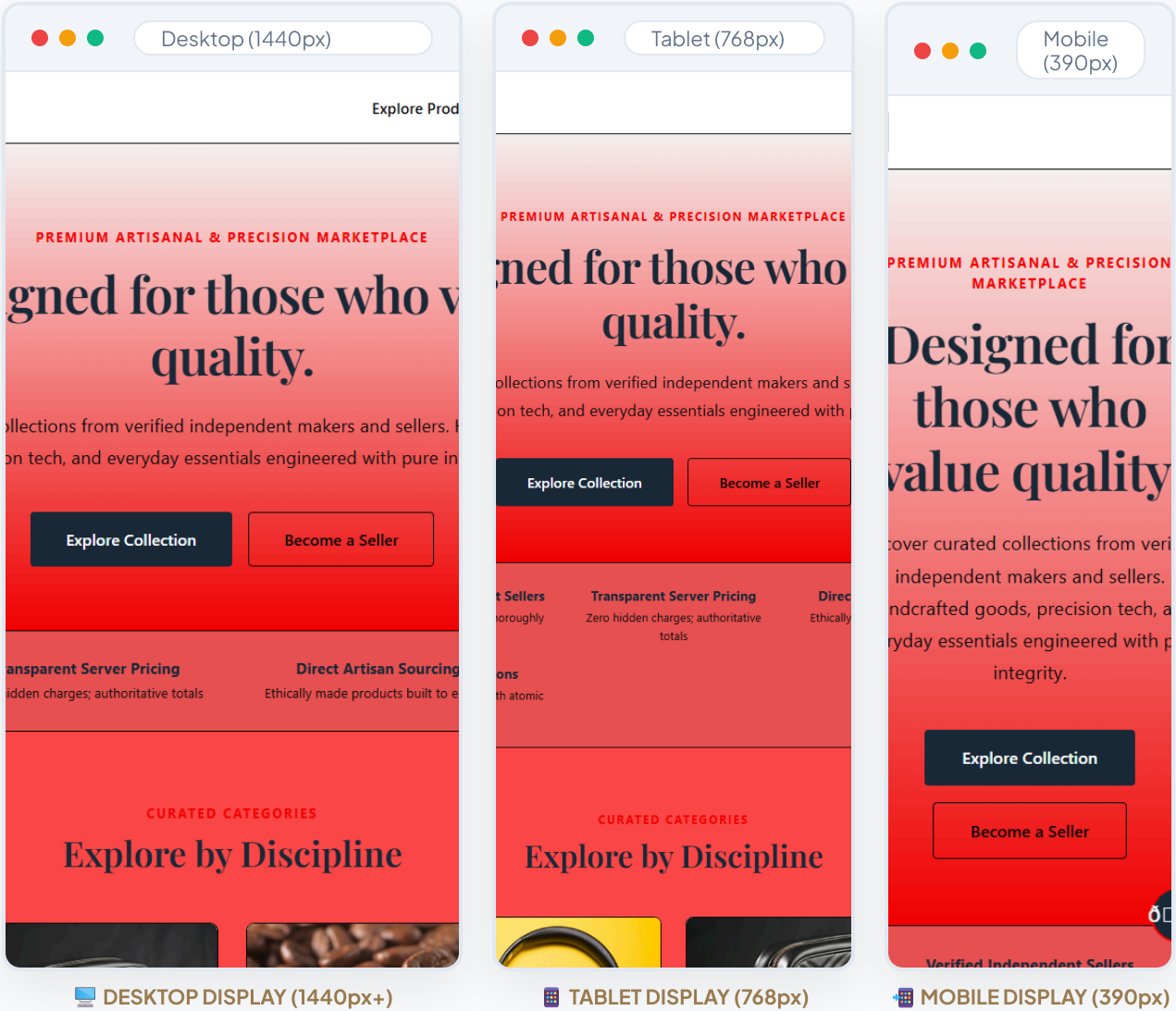
Enterprise Security

100% PreparedStatements against SQLi, CSRF synchronizer tokens, and salted BCrypt passwords.

DEVICE ADAPTABILITY

Responsive Design

Actual Live Screenshots Across Desktop, Tablet, and Mobile Screen Formats



CONSISTENT VISUAL IDENTITY: The luxury aesthetic—including gold accents, serif typography, high-res photography, and smooth transitions—is preserved uniformly whether accessed from a 4K monitor, iPad, or mobile phone.

VALUE PROPOSITION

Advantages of DJ Mart

Tangible Benefits Delivered to Shoppers and Platform Operators

**Ease of Use**

Clear layout, self-explanatory navigation, and zero prior training required for any customer.

**Saves Shopping Time**

Eliminates travel and checkout queues; complete purchases in under two minutes.

**24/7 Online Access**

Uninterrupted online storefront available anytime from anywhere with internet connectivity.

**Transparent Information**

Accurate prices, live stock status indicators, and detailed technical specifications.

**Lightweight & Blazing Fast**

Pure Vanilla JS and optimized Jakarta Servlets avoid heavy client framework bloat.

**Multi-Device Access**

Flawless responsive behavior across desktop computers, tablets, and smartphones.

**Bank-Grade Security**

Salted BCrypt hashing, CSRF synchronizer tokens, and SQL injection prevention.

**Modular & Scalable**

Layered MVC architecture enables simple integration of future enterprise modules.

STRATEGIC ROADMAP

Future Enhancements

Clearly Demarcated Features Planned for Enterprise Scale

FUTURE ROADMAP

 **Payment Gateway**

Integration with Razorpay, Stripe, and UPI for automated real-time digital payments.

FUTURE ROADMAP

 **Live Order Tracking**

Real-time parcel status tracker with courier partner integration and live updates.

FUTURE ROADMAP

 **Customer Wishlist**

Bookmark favorite luxury products for future purchases and receive price drop alerts.

FUTURE ROADMAP

 **AI Recommendations**

Machine learning models suggesting personalized products based on buyer affinities.

FUTURE ROADMAP

 **Seller Management**

Automated restocking alerts, barcode scanning, and bulk CSV catalog uploads.

FUTURE ROADMAP

 **SMS / Email Alerts**

Automated order receipts, dispatch updates, and delivery confirmations.

FUTURE ROADMAP

 **Native Mobile Apps**

Cross-platform iOS and Android apps built with Flutter for native mobile experience.

FUTURE ROADMAP

 **Advanced Analytics**

Predictive business intelligence with sales forecasting and customer churn metrics.

PROJECT SUMMARY

Conclusion

Summary of Project Achievements & Demonstration Outcome



Successful Development with Antigravity

DJ MART successfully demonstrates the rapid, robust development of a modern e-commerce web application using Google Antigravity, achieving clean architecture and enterprise code quality.



Complete End-to-End Shopping Experience

Delivered an operational shopping workflow from catalog exploration, real-time search, and category filtering to cart adjustments, address capture, and atomic order placement.



Demonstration of Modern Web Concepts

Proved the effectiveness of Jakarta EE 6 Servlets, JSTL server rendering, Vanilla JS API clients, H2 SQL database transactions, and bank-grade OWASP security protections.



Extensible & Production-Ready Foundation

The layered MVC codebase adheres to industry separation-of-concerns principles, establishing a solid foundation primed for live payment gateways and multi-cloud deployment.



THANK YOU

Questions & Answers

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